

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Computer Networks

COURSE CODE: CSA0716

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 30%

Annexure A

TECHNICAL PRESENTATION

Code of Conduct:

*I M. A Surya Kiran– 192411236(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

M. A Surya Kiran

TECHNICAL PRESENTATION

1. Edge Computing and Reliable Data Transmission Team 10 (3)

SLIDES



Introduction

Edge Computing is a computing model where data is processed near the source (such as IoT devices, sensors, cameras, or smartphones) instead of sending all data to a distant cloud server. This reduces delay (latency), saves bandwidth, and enables faster decision-making.



Edge Computing

- Processing data near the source instead of a central cloud.
- Reduces network latency.
- Saves bandwidth.
- Supports real-time decision-making.

Examples :

- ✓ Smart cities
- ✓ Self-driving cars
- ✓ Healthcare
- ✓ Smart factories



Reliable Data Transmission



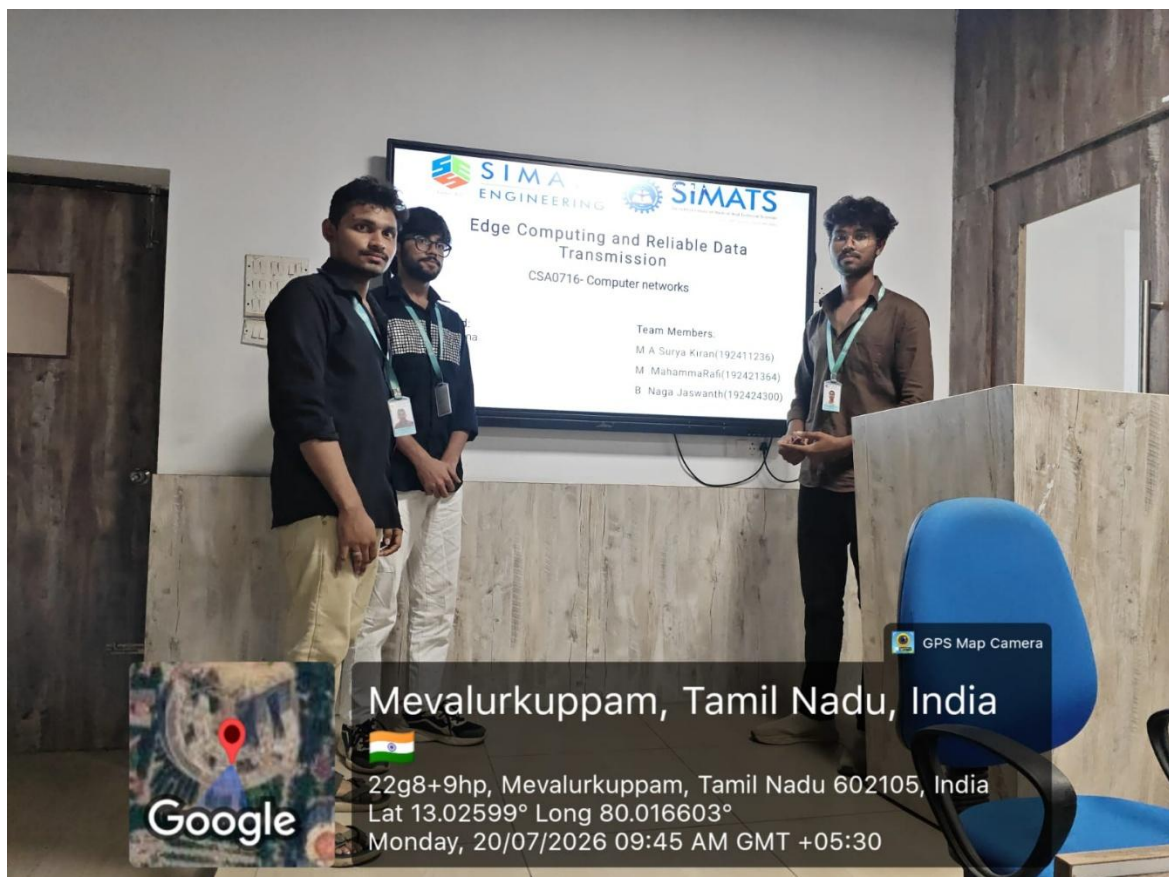
Reliable Data Transmission

Reliable Data Transmission ensures that data is delivered accurately, completely, and in the correct order from the sender to the receiver .

Key Features

- Error Detection and Correction
- Data Integrity
- Packet Acknowledgement
- Retransmission of Lost Data
- Congestion Control

GEOTAG PHOTO



RUBRICS FOR CALCULATION

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologi es	Good understandin g with proper integration of most concepts	Basic understandin g; limited or partial integration	Poor understandin g; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementati on with minor issues	Basic implementati on with errors or inefficiencies	Incorrect or incomplete implementati on
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentatio n	10	Clear, well- structured documentation with diagrams, code, and results	Organized documentatio n with necessary details	Basic documentatio n with missing details	Poor or incomplete documentatio n